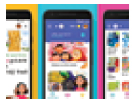
Google's Game controller
Google's patent had revealed a new Google game controller, which turned out to be the Stadia controller. **https://dgit.in/apr19-88**



Google launches Bolo App for India
Bolo is designed to help children read Hindi and English with the help of a friendly offline voice assistant. **https://dgit.in/apr19-90**

What's in a Smartphone?

While there are hundreds of brands selling smartphones, only a small cluster of companies provide the components that make them up.

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There are only a handful of companies around the world that design and fabricate the components that go into most of the smartphones everyone in the world uses. Usually, it is just one company that dominates the market share for a single component. While a majority of OEMs use 'off the shelf' components, some few manufacturers choose to use their own parts, such as SoCs or image sensors. Despite the brand on the surface, if you take a peek inside, all smartphones are more or less the same. What makes them different is how the components are put together, and which parts are chosen by the smartphone brands. Over the years, almost all the components are getting smaller, and getting placed more densely together, to make space for the one component that everyone continuously requires more from - the battery. These are the major components that go in all smartphones, and the companies that produce them.

PROCESSOR

All smartphone processors use a system on a chip (SoC) approach, which is an integrated circuit (IC) with all the most important components of the smartphone. These SoCs typically have the CPU, the GPU, a modem for communications, the GPS module, and image signal processing chips to make sense of the information captured by the camera. The CPU design itself is dominated by ARM, in collaboration with the companies that produce the different SoCs. The instruction set architecture (ISA) for nearly all smartphones are provided by ARM. In Apple smartphones, the A12 SoC is designed by Apple, but uses the ARM ISA. The Exynos chips in Samsung and the Kirin chips in Huawei devices are both designed in collaboration with ARM. The Snapdragon SOCs are designed by Qualcomm and ARM, while the Helio SoCs are designed by MediaTek and ARM. That covers nearly all, if not all the smartphones